

Belt and Suspenders: A Study of Resilience in TikTok's Global Video Delivery

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Video platforms now dominate global Internet traffic, and their performance hinges on the reliability of large content delivery network (CDN) and hosting infrastructures. Even infrequent failures can disrupt millions of users, yet little is known about how major video services engineer resilience in practice. We study TikTok, a platform whose short-form, swipe-driven workload has become a defining source of modern video traffic. Using a global active-measurement campaign across web and mobile clients, with vantage points in 30 countries and emulator-based probes in five regions, we characterize TikTok's delivery architecture and directly probe its behavior under failure.

We find that TikTok employs a multi-CDN strategy with strong regional structure, but that resilience is highly workload-dependent. Video-on-demand (VoD) delivery typically absorbs failures through structured inter- and intra-CDN fallback, while latency-critical live streaming operates under tighter constraints and, in some regions, exhibits brittle failover despite nominal provider diversity. These results provide the first empirical view into the operational logic behind TikTok's video delivery and highlight the limits of multi-CDN availability as a proxy for resilience in modern video platforms.

CCS Concepts: • **Networks** → **Network measurement**; **Network experimentation**.

Additional Key Words and Phrases: Video delivery, Resilience, Multi-CDN

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1 Introduction

Video platforms are now the single largest contributors to global Internet traffic. In most regions, video already accounts for well over half of all downstream bandwidth consumption, and the share continues to rise [33]. This growth has reshaped the delivery stack: hyperscale video applications now lean on sprawling CDN and hosting infrastructures whose performance and reliability determine whether users stay engaged or swipe away.

Yet these infrastructures fail more often than we would like to admit. Outages at major CDNs, Domain Name System (DNS) providers, and cloud regions routinely produce noticeable disruptions [14, 45–47]. CDN failures may be rare, but video streaming is unforgiving: a brief outage or regional slowdown can instantly degrade the experience for millions of users [17]. For modern video platforms, these events are operationally significant. But despite the centrality of video traffic to the modern Internet, we have limited visibility into how these platforms engineer resilience at

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scale. Understanding their robustness requires observing the delivery decisions real systems make across geographies, providers, and failure scenarios.

We focus this study on TikTok, a platform that now rivals the largest video services in reach and traffic volume. Despite its global footprint, the structure of TikTok's delivery architecture remains largely unexplored. TikTok is an especially compelling subject because its short-form, swipe-driven workload generates a volume and pattern of requests that differ sharply from conventional streaming platforms, and increasingly define modern video traffic.

While prior work has analyzed video delivery at scale, characterizing adaptive bitrate (ABR) performance, CDN selection, and quality-of-experience for platforms like YouTube and Netflix [1, 24, 32], they have focused on traditional long-form video streaming under steady-state conditions. None examine how short-form platforms like TikTok respond to delivery failures in the wild.

To study this, we conduct the first global, active measurement of TikTok's delivery system across both web and mobile access paths. We combine broad geographic coverage (containerized vantage points in 30 countries querying TikTok's web interface) with deeper, platform-specific visibility via mobile emulators deployed across five regions. This dual approach allows us to observe CDN assignment and fallback behavior at scale, independent of user data and without requiring privileged device instrumentation.

To isolate the role of the client in failure recovery, our experiments inject disruptions by blocking the client's access to the serving CDN, so the provider remains unaware of the outage. This allows us to observe the client's autonomous fallback logic, independent of any server-driven adaptation.

Our findings reveal a highly adaptive delivery architecture. TikTok selects CDNs differently across countries and network environments, and its fallback behavior under failure varies by region and traffic class. We further show that TikTok employs a multi-CDN strategy as a core component of its resilience approach, diversifying delivery paths and enabling differentiated failover behaviors. These observations suggest a resilience model shaped by regional infrastructure availability and workload constraints. Our contributions are fivefold:

- A comparative analysis of workload-sensitive resilience in TikTok, showing that buffered VoD delivery absorbs failures through structured inter- and intra-CDN fallback, while latency-critical live streaming exhibits sharply uneven and, in some regions, brittle failover behavior.
- The first large-scale empirical characterization of TikTok's multi-CDN delivery architecture, spanning web and mobile clients across 30 countries, and exposing how provider usage varies by region, client modality, and workload.
- A global, client-driven measurement methodology that reveals CDN selection and failover behavior in a black-box video platform by inducing client-local failures without triggering server-side adaptation.
- A measurement-grounded taxonomy of fallback behaviors that captures how modern video platforms operationalize resilience in practice, and explains the trade-offs between recovery speed, redundancy depth, and delay variance.
- Our measurement framework will be made available at [30] to support reproducibility and enable the community to apply our methodology to other platforms.

2 Measurement Methodology

To understand how TikTok selects and switches delivery paths, we adopt a measurement approach that captures behavior across regions, client types, and controlled disruptions. This section first outlines the methodology behind our measurement testbeds, then validates that these environments reflect real client behavior, and finally introduces the framework we use to inject failures and isolate client-side fallback logic.

2.1 Measurement Environment

To support both global breadth and platform-specific depth, we construct two complementary measurement environments. The web environment uses automated browsers routed through location-validated virtual private network (VPN) endpoints to observe TikTok's delivery behavior across 30 countries, providing wide geographic coverage and browser-level instrumentation. The mobile environment deploys Android emulators in five Amazon Web Services (AWS) regions to capture the behavior of the native client and enables controlled experiments, including induced failures. Together, these environments retrieve video sessions, expose request- and path-level metadata, and form the empirical foundation for inferring TikTok's delivery architecture.

Our measurement campaign captures steady-state delivery behavior. For the web environment, we collected data during February–May 2025 and September–November 2025; for the mobile environment, during September–December 2025. Table 3 summarizes the VoD dataset used in the workload analysis. Live measurements are summarized separately by session counts in Section 4, as they do not form a replayable video corpus with comparable per-video volume statistics. We also test temporal stability and compare overlapping countries across VPN providers, finding no evidence of provider-specific discrimination (Appendix A.5).

2.1.1 Mobile Environment. We deploy a set of mobile vantage points in five strategic AWS regions representing TikTok's major user and peering markets. Each vantage point hosts an Android emulator instantiated through the Android Virtual Device (AVD) framework [6], configured as a standard Pixel 4 running Android 11. To ensure smooth video playback and consistent network behavior, we provision dedicated bare-metal instances; this avoids the performance limitations of nested virtualization and provides stable hardware-accelerated rendering.

Within each emulator, we automate user interactions (e.g., scrolling, playback) using Appium [31]. We instrument the mobile environment with three complementary components: PcapDroid [19] provides per-app flow attribution, tcpdump records raw packet traces for timing and transport-layer analysis, and a system-wide trusted man-in-the-middle (MITM) proxy [26] exposes HTTP-level metadata for the TCP/TLS subset when decryption succeeds. Together, these components isolate TikTok traffic, measure connection setup and failover delay, and expose response headers without relying on a single observation point. For the decrypted TCP/TLS subset, we parse HTTP headers to recover delivery-path information, as these headers often expose CDN-specific cache and serving metadata. Table 4 lists the key fields. QUIC traffic is not decrypted by this pipeline; for those sessions we rely on transport metadata, Server Name Indication (SNI), DNS context, and IP/ASN inference rather than HTTP payload inspection. While media payloads remain encrypted, these signals expose CDN-specific identifiers analogous to those extracted in the web pipeline. This enables us to attribute flows to CDN providers and characterize the mobile delivery architecture under both normal operation and failure scenarios.

Vantage Point Selection. Table 2 lists the AWS regions selected to host our mobile measurement testbed. These locations provide coverage across TikTok's major user populations (North America and Europe) and topologically distinct peering regions (APAC and Oceania). This diversity allows us to observe how the delivery system behaves under substantially different latency profiles and routing conditions, without presupposing the structure of TikTok's underlying CDN deployment.

2.1.2 Web Environment. We build a complementary web measurement framework by running an instrumented Chrome browser inside a Docker container to emulate TikTok's web client. Although TikTok is mobile-first, roughly 39% of visits to TikTok.com originate from browsers [36], making the web interface an essential part of platform-wide coverage. Video playback and page interactions are automated with Selenium WebDriver [35], while the Chrome DevTools Protocol (CDP) [13] provides

Table 1. List of countries in web measurement, ranked by TikTok’s global traffic and desktop share.

Rank	Country	CC	Traffic (%)	Desktop (%)	Vantage Points
1	United States	US	22.80	36.40	17
2	Mexico	MX	3.36	34.39	4
3	Germany	DE	3.18	39.25	3
4	United Kingdom	GB	3.01	38.59	2
5	France	FR	2.79	33.28	3
6	Spain	ES	2.66	28.72	3
7	Japan	JP	2.34	32.86	3
8	Malaysia	MY	1.86	30.80	4
9	Argentina	AR	1.80	34.24	3
10	Peru	PE	1.79	48.19	3
11	Canada	CA	1.71	47.42	2
12	Poland	PL	1.27	49.92	2
13	Australia	AU	1.23	51.69	5
14	Chile	CL	0.99	36.44	1
15	South Korea	KR	0.89	42.53	1
16	Netherlands	NL	0.87	43.79	2
17	South Africa	ZA	0.75	33.56	2
18	Singapore	SG	0.64	46.69	1
19	Belgium	BE	0.53	34.42	2
20	Sweden	SE	0.41	50.61	1
21	Greece	GR	0.41	28.36	1
22	Hungary	HU	0.35	40.68	1
23	Portugal	PT	0.34	54.37	1
24	Ireland	IE	0.32	29.61	1
25	UAE	AE	0.31	43.07	3
26	Switzerland	CH	0.29	45.38	1
27	Norway	NO	0.27	50.79	1
28	New Zealand	NZ	0.25	45.20	1
29	Finland	FI	0.23	42.88	1
30	Denmark	DK	0.21	59.40	1

Table 2. List of vantage points (AWS Cloud) in mobile measurement.

Region	AWS Zone	Country	Mobile (%)
N. America	us-east-2 (Ohio)	USA	63.60
N. America	us-west-2 (Oregon)	USA	63.60
Europe	eu-west-1 (Dublin)	Ireland	70.39
Asia-Pacific	ap-SE-1 (Singapore)	Singapore	53.31
Oceania	ap-SE-2 (Sydney)	Australia	48.31

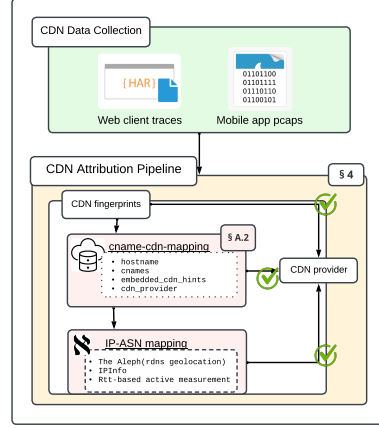


Fig. 1. Multi-layer CDN attribution pipeline combining DNS, HTTP/TLS metadata, SNI clustering, and IP/ASN inference.

HTTP-level visibility into each request. TikTok issues signed URLs for video segments; we extract these URLs and parse their response header fields (e.g., `<x-response-cache>` and `<x-cache>`), following the same header-based approach as in our mobile measurements to infer delivery-path information.

To provide global coverage, we route browser traffic through a hybrid set of vantage points spanning commercial and residential networks. For commercial VPNs (NordVPN [29], Surfshark [41]), we establish WireGuard tunnels using Gluetun [25] and attach browser containers directly to the VPN interface via Docker’s sidecar networking model. To address bias from datacenter VPN endpoints and capture traffic served from ISP-embedded caches, we complement these with residential IP endpoints from StarVPN [39]. This combination more closely resembles legitimate user connectivity across diverse network environments.

We do not extend this VPN-based location-redirection approach to the mobile setup because app-level state and instrumentation constraints make VPN-only geographic redirection unreliable in the native client; Appendix A.3 discusses this design choice in more detail.

Vantage Point Selection. For web-based measurements, we target widespread geographic coverage. We select countries based on TikTok’s web traffic rankings, prioritizing 26 of the top 50 markets identified in recent industry reports [36] that are accessible via location-validated VPNs. This ensures coverage of regions with high platform visibility. To broaden the scope, we supplement this core set with four additional countries to capture infrastructure behaviors in underrepresented

regions where CDN deployment strategies may diverge from highly connected markets. Table 1 lists the complete set of 30 countries, covering every inhabited continent¹.

Cookie-Controlled CDN Assignment. Beyond network location, TikTok uses persistent user cookies as part of its CDN selection process. During registration, the platform assigns values such as store-country-code-src and store-country, which encode the user's origin and persist across sessions. In practice, these values can override the current network location.

To isolate this effect, we systematically manipulate these cookies in controlled experiments. Using a single account, we vary country-of-origin cookie values across sessions while holding the VPN endpoint fixed. For example, accessing the platform from a French VPN while spoofing a U.S. origin results in content served from U.S. CDN nodes (e.g., New Jersey or Ashburn). Appendix A.4 provides the breakdown. This behavior indicates that TikTok's CDN selection logic reflects region-based policy constraints derived from user metadata, possibly combined with latency-aware routing.

2.2 Validation of Measurement Fidelity

We validate that our measurement environments reflect the behavior experienced by real users. First, to rule out bias from cloud or VPN-origin traffic, we compare 8,385 web sessions and approximately 1,000 mobile sessions against control measurements from residential networks and physical devices. In all cases, the delivery endpoints and response patterns observed from our controlled environments match those seen on residential clients within the same region, with no evidence of discrimination against virtualized or cloud-hosted clients (see Appendix A.5). Second, we verify that steady-state behavior remains stable before perturbation. Together, these checks give us confidence that the following sections reflect TikTok's delivery behavior under both normal and disrupted conditions.

2.3 Failure Injection Framework

Our goal in the failure experiments is to isolate how the TikTok client reacts when its active delivery path becomes unreachable. We therefore inject failures exclusively on the client side by blocking traffic to the current delivery endpoint while leaving the provider's infrastructure fully operational. This creates a controlled form of unreachability that the server cannot detect or mitigate through updated manifests, rewritten URLs, or provider-driven redirects. Any transition in delivery behavior is thus attributable to the client's own fallback logic rather than service-side adaptation.

Each experiment begins with a normal VoD or live session. We define steady state differently for VoD and live workloads. For VoD, a session becomes eligible for perturbation after initial setup completes and media arrives from a stable delivery endpoint at a consistent rate. For live, we aggregate packets into 5-tuples every 5 seconds and declare steady state only after the dominant downstream flow has remained active for at least 5 minutes while carrying the majority of downstream bytes. Once steady-state retrieval is established and the serving endpoint is stable, we inject a given failure type and monitor how the client responds across subsequent segment requests or live-frame retrievals.

VoD Workload. For VoD experiments, we curate a fixed set of 500 popular short-form videos per run for the web client, sampled from the "Explore" page. For the mobile client, which lacks this feature, we instead replay videos from the default "For You" feed of an account configured to each vantage-point region, reflecting locally popular recommended content. These videos are representative of what most users would watch: 82.5% of web and 89.2% of mobile VoD requests were served as L1 or L2 cache hits (see Appendix A.6), indicating our sampling captures the common case of TikTok's VoD delivery. Each video is played for 30–60 seconds,² long enough to observe

¹We exclude China as it utilizes the separate *Douyin* ecosystem [20].

²The average length of videos uploaded to TikTok was 42.7 seconds in 2024 [40].

multiple retrieval operations and potential changes in the delivery path. Web playback typically retrieves larger blobs over persistent connections, whereas the mobile client issues finer-grained chunk requests, offering more visibility into transitions during perturbations.

Live Workload. For live streaming, we emulate “channel surfing” behavior. The client joins a high-concurrency broadcast, remains connected for ten minutes to capture steady-state behavior, and then transitions to the next stream via a swipe or scroll gesture. This protocol is replicated for both environments across all vantage points.

Failures. The injected failures simulate abrupt loss of reachability by blocking the IP ranges associated with the active delivery endpoint using system-level firewall rules. Because the blocking occurs locally on the client, the failure prevents the client from reaching that endpoint but does not alter upstream delivery logic. We focus on the client-side reaction: (i) how quickly the client detects the stalled path, (ii) whether it retries nearby endpoints or shifts to a distinct path, and (iii) whether playback resumes smoothly or triggers rebuffering.

We intentionally target CDN-associated IP ranges at subnet granularity rather than individual servers. This does not emulate isolated server faults, which modern CDNs often absorb through internal redundancy; instead, it approximates a worst-case local loss of reachability to the serving CDN region or interconnect and exposes whether alternative delivery paths exist. Single-server failures would likely be absorbed more quickly by CDN-internal redundancy, whereas partial degradation would likely trigger different recovery behaviors; studying such cases remains future work. Before each injection, we validate rule scope against our attribution pipeline and confirm that traffic to non-targeted CDN providers remains unaffected. All failures are injected only after steady-state retrieval has been reached, ensuring that the observed responses reflect clients’ intrinsic resilience mechanisms rather than transient initialization effects.

3 CDN Attribution

Our measurements reveal that TikTok adopts a multi-CDN architecture, serving video traffic from distinct CDNs, including TikTok’s in-house platform and several third-party providers, each exhibiting characteristic DNS patterns and response-header signatures. This multi-provider architecture is not publicly documented; it emerges directly from our vantage-point observations and, as we show later, its composition is dynamic: the proportion of traffic served by each CDN changes across client modalities, regions, and even in response to failures. These observations motivate the need for a robust attribution method.

To that end, we identify the serving CDN for each request using a combination of domain naming conventions, HTTP response headers, and transport-layer metadata. These signals allow us to distinguish among major providers and track delivery-path shifts consistently across both web and mobile environments; see Fig. 1.

3.1 CDN Provider Mapping

To attribute each delivery request to its underlying CDN, we combine DNS resolution signals, HTTP-level metadata, transport-layer identifiers, and IP-based inference. Together, these signals distinguish TikTok’s in-house delivery infrastructure from the third-party CDNs observed in our measurements.

We begin by resolving full DNS canonical name (CNAME) chains for all hostnames appearing in web or mobile sessions. For raw mobile packet traces collected via tcpdump, these lookups are issued during experiment runtime from the same client vantage point that generated the corresponding requests. For offline processed PcapDroid traces, we perform the same CNAME resolution from the original emulator and network environment to preserve the client-side view. The terminal CNAME

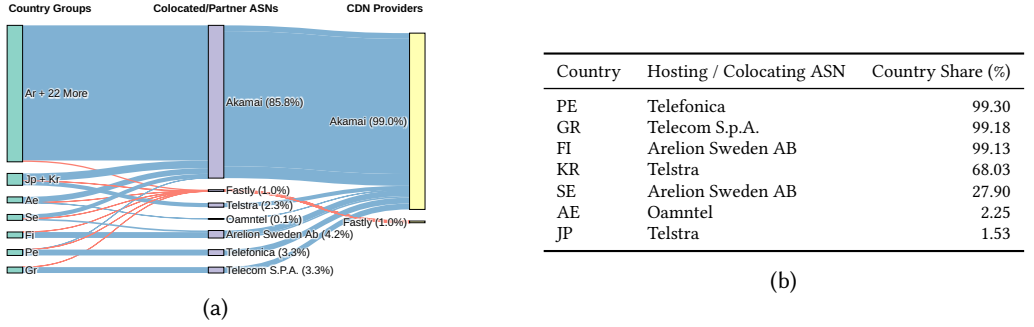


Fig. 2. (a) Web VoD CDN assignment and underlying ASN dependencies. Although delivery appears regionally diverse, most VoD traffic is served by Akamai, often via ISP-embedded or partner ASNs. (b) Akamai CDN traffic served from colocated ISP/hosting ASNs.

domains expose characteristic provider-specific patterns (e.g., Akamai edgekey domains, Fastly hostnames, or ByteDance’s `tiktokcdn.com` variants). These mappings, along with the unique embedded identifiers we extract from CNAMEs, are summarized in Appendix A.8.

For web sessions, DNS-based attribution is complemented by CDN-specific fingerprints in HTTP response headers exposed directly by the CDP, most notably the `X-Cache` field, which encodes Akamai, Fastly, and ByteDance cache semantics (details in Appendix A.8). We find that only Akamai and Fastly leave fingerprints in HTTP response headers. For mobile sessions, we use the observability signals exposed by the methodology in Section 2.1.1: decrypted TCP/TLS flows contribute HTTP metadata and SNI, while QUIC are attributed using SNI, DNS context, and IP/ASN inference. We cluster SNIs observed across CNAME-resolved domains and validated CDN IP ranges (e.g., `v45.tiktokcdn.com` → ByteDance, `v19.tiktokcdn.com` → Fastly), using these clusters as an auxiliary attribution signal.

When DNS, HTTP, and SNI evidence is insufficient, we fall back on IP-based inference. We map each delivery IP to its origin Autonomous System and geolocation using IPinfo, which reports 89% city-level accuracy within a 40 km radius [15]. To improve robustness, we cross-validate these assignments with reverse DNS analysis [44] and latency-based plausibility checks. For the latter, we compare RTTs from both the claimed geographic region and the VPN exit node; if the RTT from the claimed region is lower, we consider the geolocation plausible.

This multi-layer attribution process enables us to assign each request to a CDN provider with high confidence and to track how provider usage evolves across regions, platforms, and under failure. Fig. 14 quantifies signal coverage across our datasets. These labels form the basis for the analysis in subsequent sections.

4 Content Delivery Across Workloads

TikTok delivers two principal forms of video content, short-form Video-on-Demand (VoD) and persistent live streams, and these workloads differ sharply in cacheability, latency sensitivity, session structure, and expected traffic volume. Such differences create distinct delivery requirements, and one would expect the platform to allocate CDN providers differently across them.

To examine how TikTok assigns delivery infrastructure for each workload, we analyze a combined dataset of web and mobile sessions collected over three months. The web measurements provide geographic breadth across 30 countries (*93k VoD; 33k Live watched*), while controlled mobile experiments in five AWS regions (*7k VoD; 1k Live watched*), reveal how the native application selects providers under stable conditions. Together, these vantage points allow us to characterize

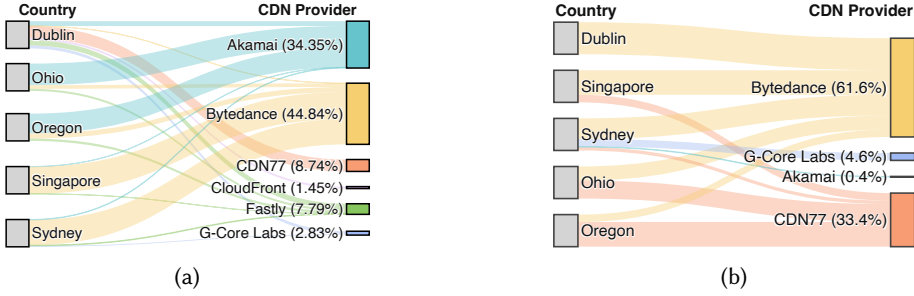


Fig. 3. Measurement overview for the mobile client. Resulting CDN assignment for mobile VoD (a) and live streaming (b), highlighting ByteDance’s dominant role in live traffic.

TikTok’s CDN usage across workloads, modalities, and regions and to identify the distinct provider allocation patterns that emerge for VoD and live streaming. Detailed findings on SNI and CDN provider mappings are provided in Table 6.

4.1 VoD CDN Selection

Web. TikTok’s VoD delivery exhibits a clear and consistent structure across regions. Fig. 2a groups countries that share similar CDN usage and Autonomous System Number (ASN) patterns based on provider assignments observed from the web client. Each source node (left) represents a country group, intermediate nodes encode the content-serving ASNs observed in our measurements, including ISP or hosting ASNs that physically host CDN caches, and the target nodes (right) represent CDN providers. Link widths reflect the normalized volume of video-segment retrievals.

Across the global footprint, Akamai dominates VoD delivery, accounting for the vast majority of observed assignments. Fastly appears in every region as a consistent but minor share, functioning as an auxiliary provider rather than a primary one. Through manual inspection in the web pipeline, we used the CDP to examine the signed segment URLs and response header fields for the Fastly subset. As detailed in Appendix A.8, the observed Content-Type values (e.g., video/mp4) indicate media delivery. Replaying unexpired signed URLs also returns playable video rather than ancillary API responses.

Regional deviations from this dominant pattern are not random. In several markets, traffic attributed to Akamai is served from distinct local ASNs. As shown in Fig. 2b, Peru routes nearly all VoD traffic through Telefónica-operated clusters; South Korea relies heavily on Telstra; Finland and Greece exhibit similar locality through Arelion and Telecom Italia Sparkle. These patterns reflect Akamai’s deeply federated “Accelerated Network Partner” (AANP) deployment model, in which serving infrastructure is embedded directly inside ISP networks [4]. In these cases, cache-related headers still identify Akamai as the CDN provider, while the serving IP addresses map to the partner network’s ASN rather than to Akamai’s canonical ASN. This strategy is evidenced by Akamai’s strategic alliances with major carriers such as Telefónica [28], Telstra [43], and Telecom Italia [42], which explicitly integrate Akamai’s delivery nodes into their respective backbone infrastructure.

As a consequence, TikTok’s apparent regional preferences partly reflect Akamai’s placement strategy rather than only explicit per-country provider decisions. Fig. 2b shows that delivery through partner and colocated ASNs is substantial in several markets. This dependency serves as an example of the “Hypergiant Off-Net” expansion [21], where delivery behavior is influenced by the physical embedding of CDN servers within access and partner networks.

Mobile. In contrast to the web client, the mobile application exhibits a stronger preference for TikTok’s proprietary infrastructure, though this strategy varies significantly by region. Aggregating

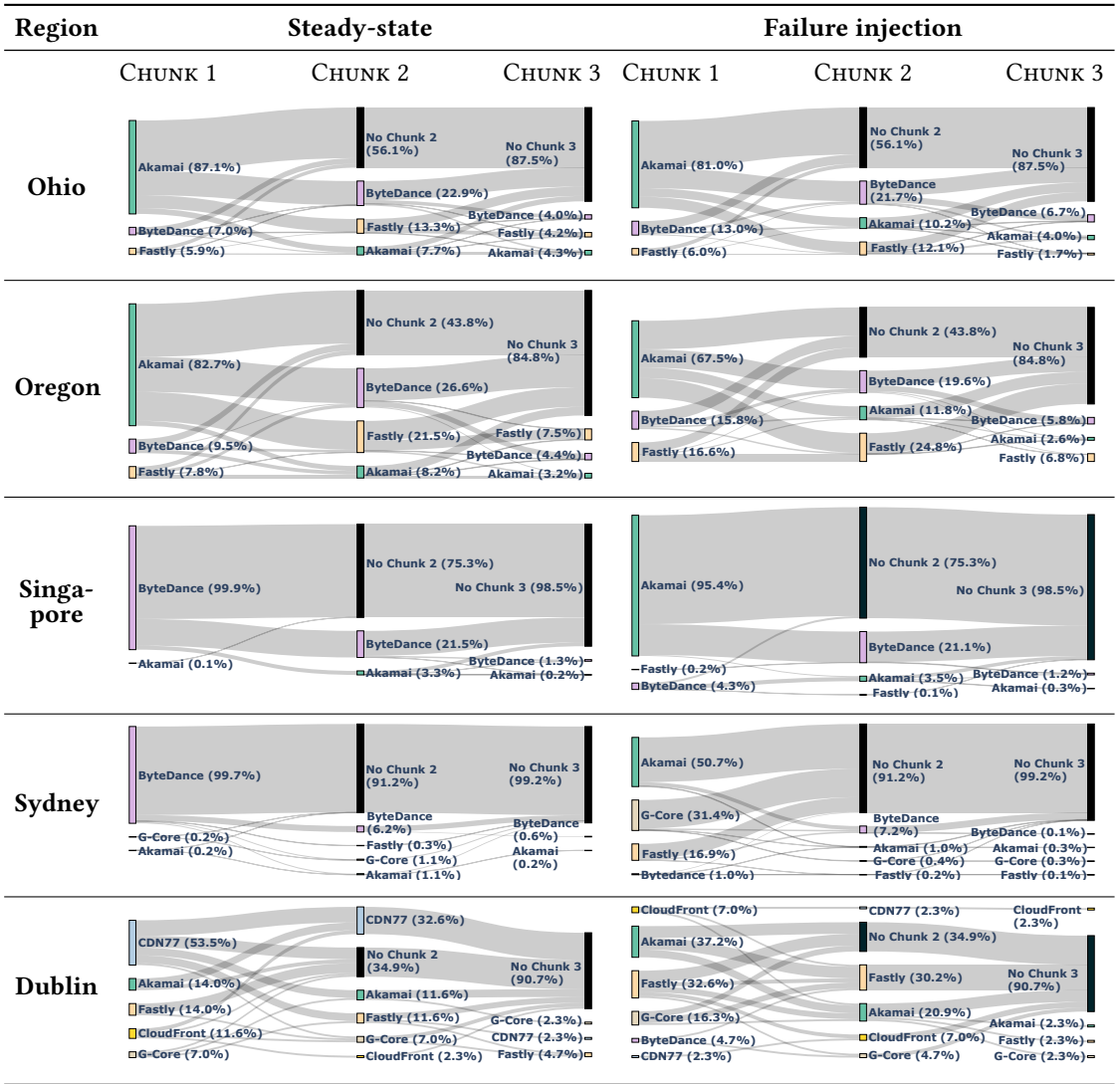


Fig. 4. Chunk-level CDN selection for mobile VoD. Startup chunks are served by broadly available CDNs, with later chunks offloaded to ByteDance or alternative providers.

video-segment bytes across playbacks, Fig. 3a shows that, when normalized across all vantage points, ByteDance is the largest single CDN provider, serving roughly 45% of the delivered payload. Akamai contributes approximately 34%, while the remaining traffic is split among CDN77, Fastly, G-Core Labs, and a small fraction from CloudFront.

The reliance on in-house delivery is most pronounced in Singapore and Sydney, where ByteDance provides 99% and 92% of video traffic, respectively. Conversely, the two vantage points in the United States rely primarily on third-party infrastructure, with Akamai serving more than 72% (Ohio) and 78% (Oregon) of their traffic. CDN distribution is most fragmented in Dublin, where ByteDance accounts for less than 1% of video delivery. Instead, the client alternates among providers led by CDN77 (47.6%), followed by Fastly (20%), Akamai (16.5%), and G-Core Labs (10%). These results

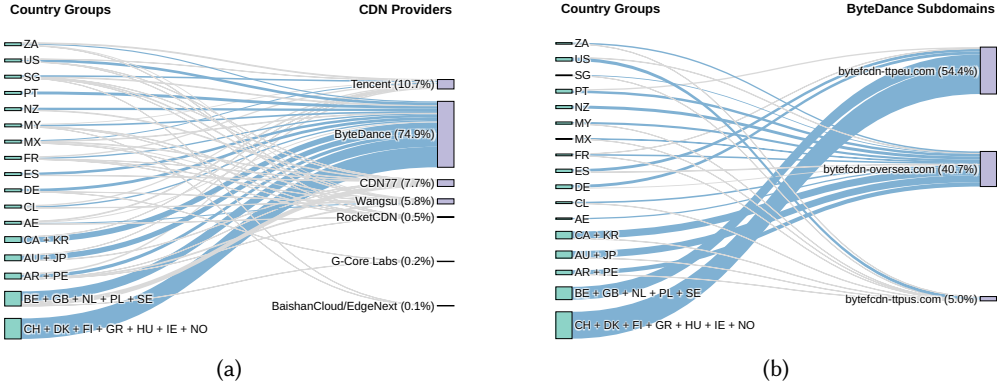


Fig. 5. Web live streaming CDN assignment. ByteDance serves most live traffic (a), organized into geographic delivery tiers with limited regional offload (b).

suggest that while TikTok prefers self-hosting, it dynamically offloads traffic to regional and global third-party CDNs where its own footprint may be less dense or cost-effective.

To understand whether CDN selection is consistent across chunks within a single video, we further analyze a subset of mobile experiment runs where video chunks were delivered over TCP/TLS and successfully decrypted by our MITM proxy, with PcapDroid used to attribute the corresponding flows to the TikTok app. Note that each run corresponds to one automation instance watching a randomized sequence of 5–10 videos; 1,422 of 1,677 runs (84.8%) across our five vantage points contained at least one eligible video, indicating broad coverage of the overall mobile dataset. In Fig. 4, the “steady-state” column provides Sankey diagrams that track the CDN provider transition for each vantage point, specifically showing the change from the initial video segment (Chunk 1) to subsequent segments (Chunk 2 and 3). Note that a significant number of flows lack segments beyond the first or second chunk (shown as “No Chunk 2” or “No Chunk 3”). This is mostly due to the video duration being too short, with occasional contributions from later segments being carried over undecryptable QUIC connections (only in a few vantage points and CDNs; see Fig. 15), or from decryption failures (e.g., certificate errors) within our MITM proxy. Hence we interpret these results as a lower bound on intra-video CDN dynamics.

Across regions, the per-chunk CDN distribution largely mirrors the aggregate traffic patterns discussed above. In Sydney and Singapore, the delivery chain is monolithic: ByteDance serves the initial chunk (>99%) and all subsequent chunks, with virtually no involvement from third-party CDNs. Dublin, by contrast, shows high fragmentation from the onset, with video segments rotating through different CDNs rather than consolidating on a single provider.

The two U.S. vantage points (Ohio and Oregon) exhibit a more nuanced, two-phase strategy. In both locations, Akamai acts as the dominant “startup” provider, serving 87.1% and 82.7% of the initial chunks, respectively. For streams that extend beyond the first segment, later chunks are more often served by alternative providers. In Ohio, the second-chunk distribution shifts toward ByteDance (22.9%) and Fastly (13.3%), with only 7.7% of second chunks remaining on Akamai. Oregon shows a similar trend, with ByteDance and Fastly gaining share in the second and third chunks. Across the two vantage points, Akamai also exhibits higher cache-hit rates than ByteDance and Fastly (see Appendix A.6 for a cache-layer breakdown and its implications for retrieval latency). Given TikTok’s swipe-based interface, where users frequently move on to the next video after only the first segment, this is consistent with a design in which highly accessed startup chunks

are placed on broadly available third-party infrastructure, while the heavier tail of video traffic is shifted to ByteDance's own CDN or other providers.

4.2 Live Streaming CDN Selection

Web. Live streaming exhibits a delivery architecture that is fundamentally different from VoD. Rather than relying on large third-party CDNs, TikTok shifts to a proprietary-first model in which ByteDance-operated infrastructure carries the bulk of global live traffic. In Fig. 5a, the majority of web sessions are served through ByteDance CDNs around 75%, supplemented by a small set of regional partners whose involvement varies by market.

Within the proprietary namespace, we observe several distinct geographic tiers. As shown in Fig. 5b, the European domain (tpeu) blends self-hosted capacity with substantial offload to CDN77, while the U.S. domain (ttpus) is served entirely from ByteDance ASNs, consistent with stricter jurisdictional controls. The broader Asia-Pacific tier (oversea) aggregates multiple partners, with Zenlayer and BEDGE as underlying ASN infrastructures (details in Appendix A.8), to support markets where ByteDance does not operate a uniform edge footprint.

Beyond these core domains, TikTok employs targeted partner offload to deepen regional coverage. Wangsu and Tencent Cloud handle meaningful shares of live traffic in Southeast Asia, and BaishanCloud/EdgeNext provides additional reach in emerging markets. This architecture reflects the demands of live workloads: real-time ingestion and low end-to-end latency, requirements poorly matched by the caching-centric hierarchies of traditional third-party CDNs.

Mobile. The mobile client follows the same high-level structure but exhibits even stronger consolidation. Fig. 3b shows that more than 61% of all mobile live sessions are served by ByteDance's own CDN infrastructure across every region we tested. Third-party providers appear less frequently than in the web traces, and when they do appear, they serve a much smaller fraction of sessions. Mobile traces also show that the client resolves more candidate CDN hostnames via DNS than it ultimately uses for content delivery, indicating a narrow preferred set of providers. In contrast, the web client distributes sessions more broadly across the resolved candidates, resulting in a more heterogeneous assignment profile.

5 Resilience and Fault Tolerance

Having characterized TikTok's steady-state multi-CDN architecture, we now turn to how the platform reacts when those delivery paths fail. We use controlled hard-failure experiments, blocking specific CDN endpoints during VoD or live sessions to observe how traffic reallocates in real time; our analysis spans the web (Section 5.2), mobile (Section 5.3), and live (Section 5.4) interfaces.

5.1 A Measurement-Grounded Fallback Taxonomy

Across thousands of failure injections, we observe that TikTok's recovery behavior falls into a small number of repeatable patterns. These patterns do not reveal internal control-plane logic, but they provide a practical, measurement-grounded vocabulary for characterizing how multi-CDN systems behave under stress.

- **Linear fallback:** the client moves sequentially to the next available CDN and does not revisit earlier options.
- **Loop retry:** the client briefly switches to an alternate provider but then re-evaluates the original primary CDN (e.g., Akamai $\rightarrow$ Fastly $\rightarrow$ Akamai), suggesting a preference hierarchy rather than a permanent handover.
- **Hierarchical fallback:** the client exhausts multiple internal options within a single provider (different ASNs or logical edge nodes) before switching to a different CDN entirely.

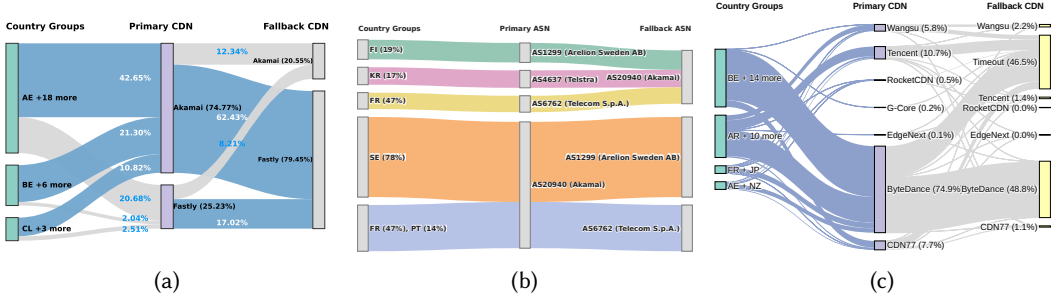


Fig. 6. Web fallback behavior under controlled failures. (a) First observed CDN-level fallback for VoD. (b) Intra-provider fallback across distinct ASNs, revealing internal redundancy within a single CDN. (c) Live streaming fallback behavior, characterized by high timeout rates and highly concentrated fallback to a small set of providers despite multiple available alternatives.

Sessions that do not recover within 10 minutes of the first injected block are classified as *timeouts*. This window spans the full post-injection period and may include multiple fallback rounds.

We classify sessions by their observed CDN transition sequence rather than by video duration. After each fallback, we block the newly selected endpoint and continue observing the same session, allowing multiple transitions within one experiment. Across 53,866 classified web VoD injections in 30 countries, we observe 29,796 linear, 7,605 loop-retry, 15,283 hierarchical, and 1,182 intra-CDN outcomes; Table 5 reports the per-country breakdown. Because classification is bounded by the 10-minute observation window, some sessions labeled linear could resolve into loop retry under longer monitoring.

These patterns appear consistently across regions and vantage points, though their prevalence varies with local infrastructure, provider diversity, and operational policy. We now examine how these behaviors manifest in each client environment.

5.2 Web VoD Fault Tolerance

On the web interface, we perform granular, in-session fault injection. Once a video stream establishes a stable delivery path, we block the serving IP subnet in real time; recovery is then governed by the browser’s TCP timeout and TikTok’s CDN-assignment logic.

Fig. 6a summarizes the first fallback observed per video. In most countries ($\approx 75\%$), Akamai serves as the primary CDN for VoD, and blocking it frequently leads to reassignment to Fastly, suggesting a structured inter-CDN fallback. In other regions seen in Fig. 6b, fallback occurs within the same provider but across distinct ASNs (e.g., Arelion Sweden AB (AS1299) switching to another Akamai ASN), indicating intra-provider redundancy.

These observations align with Section 4.1, where Akamai dominates primary delivery and Fastly appears consistently as the secondary provider. Note that Fig. 6a captures only the *first* fallback event; subsequent failures reveal multi-step fallback chains. In our traces, these CDN transitions are initiated by the client following DNS re-resolution rather than by a server-directed playlist update: each switch is preceded by new DNS lookups for alternative delivery hostnames, and we do not observe a manifest reload before the client opens connections to the replacement endpoint.

Fallback Taxonomy in the Web Environment. The fallback taxonomy in Fig. 7 varies across countries, consistent with differences in provider diversity, internal footprint, and recovery depth. These differences should not be interpreted solely as intentional per-country policy choices, as local CDN availability may also shape the observed patterns:

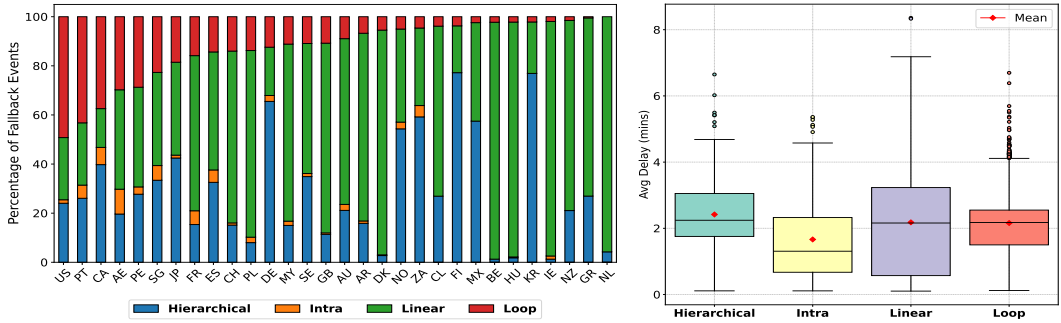


Fig. 7. Measurement-grounded fallback taxonomy and recovery cost. Distribution of fallback strategies (left) and corresponding recovery delays (right), showing that intra-CDN and hierarchical fallback recover faster than cross-provider transitions, at the cost of greater delay variance.

Linear fallback dominates globally (e.g., Netherlands, Ireland, Belgium, Brazil). Here, TikTok progresses through CDNs without revisits, consistent with stable, well-provisioned Akamai deployments that require minimal recovery complexity.

Loop retry (e.g., United States, Portugal, Canada, UAE) reflects a tiered reliability model: the primary CDN remains the “happy path,” and the secondary provider serves as a transient circuit breaker rather than a permanent handover. This behavior is consistent with volume-based pricing commitments and the high likelihood that primary-path failures are transient rather than structural.

Hierarchical fallback (e.g., Korea, Finland, Denmark, Mexico) emerges where a provider offers a large internal footprint. Clients attempt multiple logical edges or ASNs within the same CDN before escalating to a different provider, likely to preserve cache locality and avoid cross-CDN origin fetches.

Impact on Delay. For the web client, we do not observe the browser player’s effective playout buffer directly, so we interpret the delays in Fig. 7 as network-level recovery times rather than exact user-visible stalls. Linear fallback yields predictable delay, while retry-based strategies introduce higher variance and long-tail cases. These patterns suggest that TikTok trades increased delay dispersion for operational benefits such as cache reuse and reduced cross-CDN traffic.

5.3 Mobile VoD Fault Tolerance

Unlike the web client, TikTok’s mobile application does not expose individual segment requests through a debuggable browser stack, and identifying specific video content inside encrypted QUIC/TLS traffic is not feasible without heavyweight instrumentation. We therefore restrict our analysis to the subset of mobile VoD playbacks delivered over TCP/TLS and successfully intercepted by our MITM proxy, which allows us to extract HTTPS response headers (video identifiers, segment metadata, server IPs, etc.).

For each vantage point, we collect videos in batches and build a mapping of $\langle \text{video_id}, \text{IP_subnets} \rangle$ from the decrypted headers; a video may map to multiple subnets if different chunks were served from different edges. Failure injection is then applied at this granularity: before replaying a selected video, we block all subnets associated with it using system-level rules on the Android emulator. Comparing CDN and subnet choices before (steady-state) and after injection reveals how the mobile delivery stack reacts when specific serving edges become unavailable. Because we operate at the subnet level rather than disabling an entire CDN, the same provider may reappear under failure via a different unblocked edge.

Traffic Redistribution. In Fig. 8a, we show CDN usage after failure injection across all vantage points. In Singapore and Sydney, displaced traffic is absorbed primarily by Akamai (87% and 49%, respectively). Dublin remains fragmented, with CDN77 carrying 40% and the remainder split among Akamai, Fastly, CloudFront, and G-Core Labs. Notably, ByteDance’s share remains below 1% even under failure injection. In the U.S., failover is distributed: in Oregon, Akamai’s share drops from 72.5% to 46.9%, with ByteDance rising from 18.7% to 40.5% and Fastly from 8.8% to 12.6%.

Chunk-Level Failover Behavior. Fig. 4 compares chunk-level CDN transitions before and after subnet blocking. In Singapore and Sydney, where steady-state delivery relies almost exclusively on ByteDance, failure injection shifts the *first* chunk to third-party CDNs (e.g., Akamai for 95.4% of videos in Singapore), but later chunks revert to ByteDance. This suggests third-party CDNs function primarily as startup bootstrappers, mitigating initial stall risk before the session settles back onto ByteDance’s infrastructure.

The U.S. vantage points exhibit a different pattern: most videos still fetch their first chunk from Akamai (81% in Ohio, 67.5% in Oregon), but from edges different from steady-state. Subsequent chunks show little change, indicating that Akamai’s internal redundancy absorbs the failure, with no need to escalate to another CDN. Hence the broader delivery behavior (startup on Akamai, bulk transfer via other CDNs) remains intact.

Fallback Strategies. We apply the same fallback taxonomy from Section 5.1 at the chunk level. For each video, we run two rounds of failure injection: (i) replay while blocking steady-state subnets, then (ii) replay again while blocking the union of subnets observed across the first two runs. We classify the resulting provider sequences as: *linear fallback*, *loop retry*, *hierarchical fallback*, or *intra-CDN rerouting* (all chunks remain within one CDN).

Fig. 8b shows that, in the U.S., intra-CDN rerouting is most common (43.4% in Ohio, 47.9% in Oregon), highlighting strong subnet-level redundancy within Akamai. The remaining sequences split across loop retry, hierarchical fallback, and linear fallback. Singapore is dominated by hierarchical fallback (92.6%), reflecting extensive intra-provider exploration before switching CDNs. Sydney, in contrast, is primarily linear (69.9%), indicating more direct progression across distinct providers. We do not report taxonomy results for Dublin due to insufficient decryptable sequences across all three runs. As with the web taxonomy, classification is bounded by our observation window, so a small fraction of sequences labeled linear may resolve into loop retry under longer replay.

Impact on Delay. Recovery delay for TCP/TLS chunks is shown in Fig. 8c. These delays are measured in the network trace, but prior work helps interpret their playback relevance: TikTok fetches chunks on demand rather than preloading full videos, and aggressive prefetching is costly in swipe-driven viewing [18, 23]. Using our measured chunk sizes, Dashlet’s observation that TikTok prefetches the first chunks of the next five videos, and a nominal playback rate of 1 Mbps, we estimate total buffered content on the order of 10 s. While the time-sensitive mobile setting generally resolves failures more quickly than the browser-based web environment (Fig. 7), trends among fallback behaviors remain consistent. Intra-CDN rerouting yields the shortest and most predictable recovery times, while cross-provider transitions (especially linear fallback) produce higher and more variable delays. These results apply only to the observable subset of mobile traffic (TCP/TLS chunks), but they broadly mirror the trade-offs seen in the web interface.

5.4 Web Live Fault Tolerance

Live streaming operates with minimal playout buffers to limit end-to-end latency for real-time delivery. As a result, any disruption in the delivery path is exposed immediately to the client.

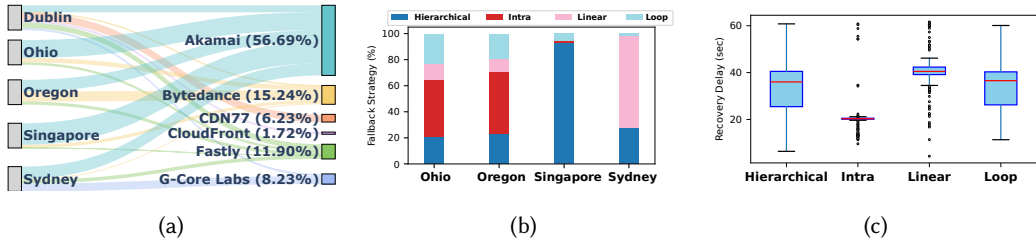


Fig. 8. CDN distribution for video chunks (a), chunk-level failover behaviors distribution (b), and recovery delay distribution (c) for mobile VoD under failure injection.

We apply the same recursive CDN blocking methodology used for VoD (Section 5.2) to active live sessions. After we block the active delivery path, the client issues additional DNS lookups and attempts replacement delivery flows to newly resolved endpoints. When recovery succeeds, we observe a corresponding hostname or IP-subnet change followed by resumed downstream media traffic. As shown in Fig. 6c, the steady-state ecosystem for live traffic is more diverse than for VoD but remains heavily consolidated: ByteDance serves 74.9% of sessions, followed by Tencent (10.7%) and Wangsu (5.8%).

Failure injection reveals two broad outcomes. First, we observe a high incidence of timeouts: 46.5% of fallback attempts fail to recover within the client-visible timeout window, reflecting the limited buffering available during live playback. Second, for sessions that do recover, distinct patterns emerge. ByteDance tends to handle failures internally (intra-CDN rerouting), while third-party providers show a characteristic escalation: when Tencent or Wangsu fail, a portion of sessions successfully fall back to ByteDance rather than retrying within the same provider. This behavior suggests that ByteDance's infrastructure functions as a universal fallback path for live traffic, acting as a resilience anchor when external CDNs cannot meet latency or availability requirements.

5.5 Mobile Live Fault Tolerance

We now examine live streaming on the mobile client, focusing on regions where live traffic volume and operational significance warrant deeper analysis.

Our active failure injections reveal substantial regional variation in resilience. In the United States, the system adopts a dual-primary configuration: CDN77 (56.2%) and ByteDance (43.8%) serve as near-equal entry points. As shown in Fig. 9a, blocking either provider redirects traffic cleanly to the other, with negligible timeout rates at the first fallback tier, indicating that the deployment maintains sufficient redundant capacity to absorb full load shifts between the two.

In contrast, several Asia-Pacific regions exhibit what we term *brittle diversity*. Although these markets use multiple providers in steady state, this diversity does not translate into effective resilience under failure. Quantitatively, first-tier timeout rates range from negligible in the U.S. to 43.8% in Singapore and 81.0% in Australia. In Australia, for example, live traffic is distributed across ByteDance (63.0%), G-Core Labs (24.0%), CDN77, and Akamai. Yet blocking the primary provider leads to an 81.0% timeout rate, indicating that the system cannot coordinate a timely fallback before playback drains its minimal buffer. Singapore shows a similar pattern, with a 43.8% initial timeout rate despite nominal reliance on ByteDance and CDN77. In regions without effective cross-CDN fallback, these timeout rates are consistent with both our coarse blocking model and the limited recovery diversity exposed by the traces, indicating weaker resilience under this class of failure.

These results highlight a key distinction from VoD: for latency-sensitive workloads, *the existence of multiple providers is not equivalent to resilience*. Only deployments with tightly integrated fallback

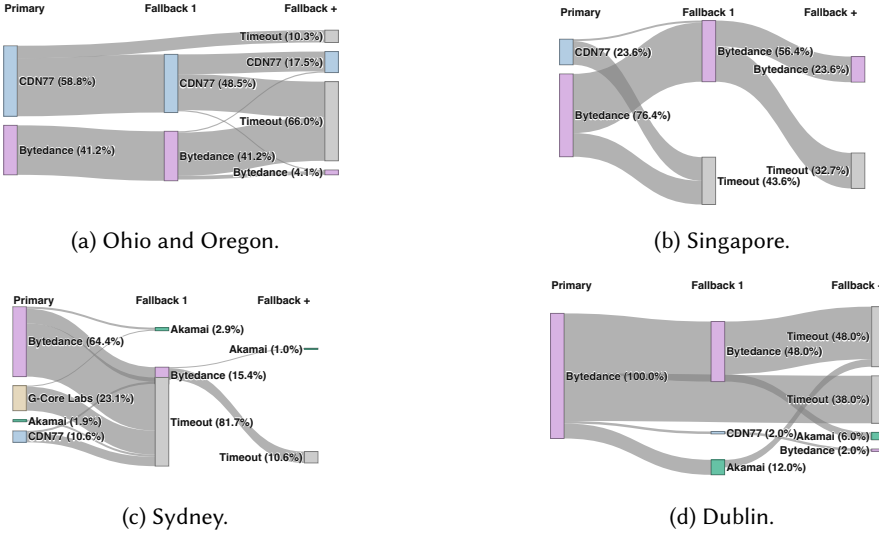


Fig. 9. Live streaming failover patterns across five regions. CDN traffic redistribution when recursively blocking CDNs in (a) Ohio and Oregon, (b) Singapore, (c) Sydney and (d) Dublin.

paths, such as the U.S., maintain service continuity under stress. In regions without this integration, live failover remains highly fragile despite apparent provider diversity.

We do not report recovery delay for live streaming because the majority of live failovers do not complete successfully: the first fallback attempt often exceeds the tight latency budget of live playback, resulting in an immediate timeout rather than a measurable recovery interval.

Across all environments, TikTok’s response to failure reflects a common set of recovery patterns but sharply different levels of robustness. VoD sessions on both web and mobile generally withstand failures through structured inter- and intra-CDN fallback, with recovery shaped by local provider diversity and the depth of redundancy within Akamai and ByteDance’s deployments. Live streaming, by contrast, exposes the limits of this architecture: the absence of meaningful buffering turns many failures into immediate playback stalls, and only regions with tightly integrated provider pairs (e.g., the U.S.) sustain reliable recovery. Taken together, these results show that TikTok’s multi-CDN design provides broad resilience for throughput-oriented workloads, but latency-critical live traffic depends on far narrower and more regionally uneven failover pathways.

6 Discussion and Limitations

Our observations capture how TikTok’s delivery system behaves across diverse networks and failure conditions. We first highlight the broader implications of these findings beyond TikTok, and then outline the main limitations that bound our interpretation.

6.1 Broader Implications

Although our study focuses on TikTok, the implications extend beyond a single platform. The fallback taxonomy provides a reusable framework for characterizing multi-CDN resilience in other large-scale services. More broadly, our results show that provider diversity alone does not guarantee resilience: the brittle-diversity behavior we observe in some regions indicates that redundancy also depends on how fallback paths are integrated in operation. Our methodology can also help operators and researchers reason about traffic shifts during CDN disruptions. While TikTok’s CDN

assignments may evolve, the measurement framework, client-local failure injection method, and fallback taxonomy are not tied to a single deployment snapshot.

6.2 Limitations and Scope

We next outline the main methodological limitations and their implications for interpretation.

Encrypted transport and partial mobile observability. On mobile, encryption restricts per-request visibility to sessions where our controlled devices expose headers through MITM. While this limits granularity, the fallback patterns we study are coarse behaviors (provider shifts, path changes, and timing signals) that remain observable even when finer request context is opaque. A natural next step is attribution methods that operate robustly over QUIC-only telemetry.

Geographic and network coverage. Our vantage points span the regions that account for the majority of TikTok's global traffic footprint and include both datacenter and residential access paths. While we do not capture every ISP- or country-specific configuration, the behaviors we analyze reflect global mechanisms rather than local exceptions. Future studies could extend this approach to long-tail regions or mobile-carrier-specific deployments.

VPN-induced latency. Our VPN-based analysis is limited to web sessions; although Appendix A.5 shows that residential and VPN-based web sessions expose nearly identical CDN choices, VPN tunnels can still introduce additional absolute latency. We therefore interpret fallback behavior primarily through relative differences rather than absolute delay values.

Controlled-account behavior. Our measurements use a small number of researcher-controlled accounts to ensure reproducibility and controlled manipulation of session metadata (e.g., cookies). This abstracts away personalization effects that accumulate on real user accounts, and personalization could in principle influence delivery behavior. However, we observe consistent CDN assignment patterns across sessions, devices, and time periods within the same geographic region, suggesting that CDN selection is driven primarily by geographic and infrastructure factors rather than by account-level personalization. Future work could examine how personalization signals interact with CDN assignment or fallback logic more directly.

Client-local failure model. Our failure injections operate solely at the client's view of the network, allowing us to isolate the platform's fallback behavior without inducing real outages. This approach is safe and repeatable, but it does not capture system-wide adaptations during large-scale provider or POP failures. Cooperative experiments or natural outages would provide a complementary perspective.

Inference scope and taxonomy. Our conclusions about TikTok's operational logic are necessarily indirect: they are inferred from observable network behavior rather than from access to TikTok's internal orchestration systems or policies. The fallback taxonomy therefore characterizes repeatable external behaviors, not TikTok's internal algorithms, and alternative explanations may exist for some patterns we observe, including transient capacity constraints, contractual routing preferences, or internal cost optimization. We interpret these explanations as hypotheses consistent with the data, not as confirmed mechanisms. The Sankey diagrams complement, rather than replace, the quantitative analysis in the main text and appendix. As attribution techniques and vantage coverage improve, the taxonomy can evolve to capture finer distinctions or new behaviors.

7 Related Work

To meet growing user expectations for high-quality, low-latency video streaming, content providers have largely abandoned single-CDN dependencies in favor of multi-CDN strategies [8, 9, 38]. By dynamically leveraging multiple providers (e.g., Akamai, Fastly, Cloudflare), platforms can optimize for regional performance, reduce vendor lock-in, and ensure fault tolerance [5, 37]. Prior literature has extensively modeled these selection strategies, proposing algorithms that balance cost, latency,

and throughput using both DNS-based and client-side switching mechanisms [7, 22]. However, most of these studies rely on simulations or controlled testbeds rather than large-scale measurements of proprietary, production-grade logic used by hyper-scale platforms.

CDN Measurement and Analysis. A substantial body of work has characterized the performance of individual CDNs [1, 11] and their mapping behaviors [12]. More recently, researchers have focused on understanding the delivery infrastructures of specific video giants, such as Netflix's Open Connect [10], YouTube's cache hierarchy [3], and Hulu's delivery ecosystem [2].

In the context of ByteDance/TikTok, recent internal studies have detailed the design of hybrid content delivery systems that integrate traditional CDNs with peer-to-peer (P2P) distribution to optimize bandwidth costs [49, 50]. While these first-party publications provide valuable insights into the platform's architectural goals, they primarily offer idealized views and lack independent verification of how the system performs under the unconstrained and chaotic conditions of the public Internet. Despite the platform's massive global footprint, its content delivery patterns and dynamic multi-CDN decision logic remain largely opaque to the networking research community.

Resilience and Failover Mechanisms. Resilience is a primary motivator for multi-CDN adoption. Standard failover mechanisms typically involve DNS-based load balancing, where traffic is shifted at the resolution level during outages, or client-side logic, where the player actively switches hosts upon detecting stalling or HTTP errors [27]. Recent work has highlighted the complexity of "seamless" switching, noting that aggressive failover policies can inadvertently cause route flapping or cache thrashing [38]. Understanding how latency-sensitive applications like TikTok, which requires near-instant video start times manage these transitions during provider failures is critical for advancing the state of robust video delivery.

Unlike prior studies that focus on single-provider performance [51] or internal design documents [34], we utilize distributed active probing to reveal how TikTok effectively selects, falls back, and transitions across providers, offering new insights into the resilience of modern short-video delivery ecosystems.

8 Conclusion

We presented a large-scale empirical study of TikTok's multi-CDN delivery system that moves beyond steady-state observation to directly probe resilience under failure. By inducing client-local disruptions across web and mobile vantage points worldwide, we expose adaptation logic that is otherwise hidden behind CDN abstractions.

Our measurements show that resilience in modern video platforms is strongly workload-dependent. TikTok's VoD delivery typically absorbs failures through structured inter- and intra-CDN fallback, while latency-critical live streaming operates under far tighter constraints and, in some regions, exhibits brittle failover behavior. The contrast between symmetric recovery in the U.S. and nominal but ineffective diversity in parts of APAC demonstrates that multi-CDN availability alone does not guarantee redundancy.

Together, these results highlight the need to reason about resilience in terms of workload, latency budget, and integration depth, rather than provider count alone.

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References

- [1] Vijay Kumar Adhikari, Yang Guo, Fang Hao, Volker Hilt, and Zhi-Li Zhang. 2012. A Tale of Three CDNs: An Active Measurement Study of Hulu and Its CDNs. In *Proc. of IEEE INFOCOM Workshops*.
- [2] Vijay Kumar Adhikari, Yang Guo, Fang Hao, Volker Hilt, and Zhi-Li Zhang. 2012. Unreeling Netflix: Understanding and Improving Multi-CDN Movie Delivery. In *Proc. of IEEE INFOCOM*.
- [3] Vijay Kumar Adhikari, Sourabh Jain, Yingying Chen, and Zhi-Li Zhang. 2012. Vivisecting YouTube: An Active Measurement Study. In *2012 Proceedings IEEE INFOCOM*. 2521–2525. doi:10.1109/INFOCOM.2012.6195644
- [4] Akamai Technologies. 2024. Akamai Accelerated Network Partner (AANP) Program: Executive Summary. <https://www.akamai.com/us/en/multimedia/documents/brochure/akamai-accelerated-network-partner-program-executive-summary.pdf>. Accessed: 2025-12-11.
- [5] Ali Al-Shammari and Wei Zhang. 2023. Vendor Management in Multi-Cloud Ecosystems: Avoiding Lock-In and Optimizing Partnerships. In *Proc. of CLOSER*.
- [6] Android Developers. 2025. Android Virtual Device (AVD) — Android Emulator. <https://developer.android.com/studio/run/managing-avds>. Accessed: 2025-11-27.
- [7] Abdelhak Bentaleb, Ali C. Begen, and Roger Zimmermann. 2019. A Survey on Bitrate Adaptation Schemes for Streaming Media Over HTTP. *IEEE Communications Surveys & Tutorials* 21, 1 (2019), 562–585.
- [8] Abdelhak Bentaleb, Reza Farahani, Farzad Tashtarian, Hermann Hellwagner, and Roger Zimmermann. 2023. Which CDN to Download From? A Client and Server Strategies. In *Proc. of MHV*.
- [9] Jeremias Blendin, Fabrice Bendfeldt, Ingmar Poesche, Boris Koldehofe, and Oliver Hohlfeld. 2018. Dissecting Apple's Meta-CDN during an iOS Update. In *Proc. of IMC*.
- [10] Timm Bottger, Felix Cuadrado, Gareth Tyson, Ignacio Castro, and Steve Uhlig. 2016. Open Connect Everywhere: A Glimpse at the Internet Ecosystem through the Lens of the Netflix CDN. In *Proc. of IMC*.
- [11] Matt Calder, Ashley Flavel, Ethan Katz-Bassett, Ratul Mahajan, and Jitendra Padhye. 2015. Analyzing the Performance of an Anycast CDN. In *Proc. of IMC*.
- [12] Fangfei Chen, Ramesh K. Sitaraman, and Marcelo Torres. 2015. End-User Mapping: Next Generation Request Routing for Content Delivery. In *Proc. of ACM SIGCOMM*.
- [13] Chromium Project. 2024. Chrome DevTools Protocol. <https://chromedevtools.dev/>.
- [14] Cloudflare. 2025. Cloudflare outage on November 18, 2025. <https://blog.cloudflare.com/18-november-2025-outage/>. Accessed: 2025-12-10.
- [15] Omar Darwich, Hugo Rimlinger, Milo Dreyfus, Matthieu Gouel, and Kevin Vermeulen. 2023. Replication: Towards a Publicly Available Internet Scale IP Geolocation Dataset. In *Proc. of IMC*.
- [16] David Dittrich and Erin Kenneally. 2012. The Menlo Report: Ethical Principles Guiding Information and Communication Technology Research. *US Department of Homeland Security* (2012).
- [17] Florin Dobrian, Vyas Sekar, Asad Awan, Ion Stoica, Dilip Joseph, Aditya Ganjam, Jibin Zhan, and Hui Zhang. 2011. Understanding the Impact of Video Quality on User Engagement. In *Proc. of ACM SIGCOMM*.
- [18] Jinlong E, Wei Xu, Jianfei Bi, Lin He, Haoyang Li, Anqi Gu, Dan Yang, and Yunpeng Chai. 2025. Wisely Optimizing Short Video Streaming for a User-Vendor Win-Win Outcome. In *Proceedings of the ACM SIGCOMM 2025 Conference* (São Francisco Convent, Coimbra, Portugal) (SIGCOMM '25). New York, NY, USA, 1272–1274. <https://doi.org/10.1145/3718958.3754346>
- [19] Emanuele Faranda. 2024. PCAPdroid: Network Monitor and Capture Tool for Android. <https://github.com/emanuele-f/PCAPdroid>. Accessed: 2024-05-20.
- [20] Claire Fu and Daisuke Wakabayashi. 2024. There Is No TikTok in China, but There Is Douyin. Here's What It Is. <https://www.nytimes.com/2024/04/25/business/china-tiktok-douyin.html>. Accessed May 10, 2025.
- [21] Petros Gigis, Matt Calder, Lefteris Manassakis, George Nomikos, Vasileios Kotronis, Xenofontas Dimitropoulos, Ethan Katz-Bassett, and Georgios Smaragdakis. 2021. Seven Years in the Life of Hypergiants' Off-Nets. In *Proc. of ACM SIGCOMM*.
- [22] Burak Kara and Gwendal Simon. 2024. Power Efficient Multi-CDN Communication over Content Steering Server. In *Proc. of MMSys*.
- [23] Zhuqi Li, Yaxiong Xie, Ravi Netravali, and Kyle Jamieson. 2023. Dashlet: Taming Swipe Uncertainty for Robust Short Video Streaming. In *20th USENIX Symposium on Networked Systems Design and Implementation (NSDI 23)*. USENIX Association, Boston, MA, 1583–1599. <https://www.usenix.org/conference/nsdi23/presentation/li-zhuqi>
- [24] Hongzi Mao, Ravi Netravali, and Mohammad Alizadeh. 2017. Neural adaptive video streaming with pensieve. In *Proc. of ACM SIGCOMM*.
- [25] Quentin McGaw. 2024. Gluetun: VPN Client in a Thin Docker Container for Multiple Providers. <https://github.com/qdm12/gluetun>. Supports WireGuard and OpenVPN.
- [26] mitmproxy. 2025. mitmproxy: An Interactive HTTPS Proxy. <https://mitmproxy.org/>.

- [27] Matthew K. Mukerjee, David Naylor, Junchen Jiang, Dongsu Han, Srinivasan Seshan, and Hui Zhang. 2015. Practical, Real-Time Centralized Control for CDN-Based Live Video Delivery. In *Proc. of SIGCOMM*.
- [28] Netify. 2025. Netify. <https://www.netify.ai/resources/telco/telefonica/country/pe> Accessed May 15, 2025.
- [29] NordVPN. 2024. NordVPN: Secure and Private Access to the Internet. <https://nordvpn.com/>.
- [30] Northwestern University AquaLab. 2024. Multi-CDN Resilience Measurement. <https://github.com/NU-AquaLab/multi-cdn-resilience-measurement>. GitHub repository, accessed April 2026.
- [31] OpenJS Foundation. 2024. Appium: Automation for iOS, Android, and Windows Apps. <http://appium.io>. Accessed: 2024-05-20.
- [32] Irena Orsolic, Dario Pevec, Mirko Suznjevic, and Lea Skorin-Kapov. 2016. QoE measurement of mobile YouTube video streaming. In *Proc. of PIMRC*.
- [33] Sandvine. 2023. *The Global Internet Phenomena Report 2023*. Technical Report. Sandvine. <https://www.sandvine.com/global-internet-phenomena-report-2023> Market Report. Video category accounts for 65.93% of total global internet traffic volume..
- [34] Brandon Schlinker, Hyojeong Kim, Timothy Cui, Ethan Katz-Bassett, Harsha V Madhyastha, Italo Cunha, James Quinn, Saif Hasan, Petr Lapukhov, and Hongyi Zeng. 2017. Engineering egress with edge fabric: Steering oceans of content to the world. In *Proc. of ACM SIGCOMM*. 418–431. doi:10.1145/3098822.3098853
- [35] Selenium Project. 2024. Selenium WebDriver. <https://www.selenium.dev/>.
- [36] SEMrush. 2025. TikTok.com Traffic Analytics and Market Share. <https://www.semrush.com/website/tiktok.com/overview/> Accessed April 29, 2025.
- [37] Jashanjot Singh Sidhu, Chidambar Joshi, and Abdelhak Bentaleb. 2025. A Multi-CDN Playground for Dash.js: Enabling Integration of CDN Switching Strategies. In *Proc. of MMSys*.
- [38] Daniel Silhavy, Will Law, Stefan Pham, Ali C. Begen, Alex Giladi, and Alex Balk. 2023. Dynamic CDN Switching – DASH-IF Content Steering in dash.js. In *Proc. of MHV*.
- [39] StarVPN App. 2024. StarVPN: Residential IP VPN Network. <https://www.starvpn.com/>. Accessed: 2025-12-08.
- [40] Statista. 2025. Average TikTok video duration worldwide from 2023 to 2024. <https://www.statista.com/statistics/1485205/tiktok-video-duration/>. Accessed: 2025-12-12.
- [41] Surfshark B.V. 2024. Surfshark: Secure VPN for Privacy and Security. <https://surfshark.com>. Accessed: 2024-05-20.
- [42] Telecom Italia Sparkle S.p.A. 2025. Telecom Italia Sparkle S.p.A. <https://www.netify.ai/resources/telco/gruppo-tim> Accessed Nov 12, 2025.
- [43] Telstra. 2025. Telstra. <https://www.telstrainternational.com/en/wholesale> Accessed May 15, 2025.
- [44] Kedar Thiagarajan, Esteban Carisimo, and Fabián E. Bustamante. 2025. The Aleph: Decoding Geographic Information from DNS PTR Records Using Large Language Models. In *Proc. of CoNEXT*.
- [45] ThousandEyes Internet Research Team. 2025. AWS Outage Analysis: October 20, 2025. <https://www.thousandeyes.com/blog/aws-outage-analysis-october-20-2025> Accessed: 2025-12-10.
- [46] ThousandEyes Internet Research Team. 2025. Google Cloud Outage Analysis: June 12, 2025. <https://www.thousandeyes.com/blog/google-cloud-outage-analysis-june-12-2025> Accessed: 2025-12-10.
- [47] ThousandEyes Internet Research Team. 2025. Microsoft Azure Front Door Outage Analysis: October 29, 2025. <https://www.thousandeyes.com/blog/microsoft-azure-front-door-outage-analysis-october-29-2025> Accessed: 2025-12-10.
- [48] ToDetect. 2026. TikTok Timezone Analysis and Detection. <https://www.todetect.net/article/timezone/tiktok-timezone>. Online article, accessed April 2026.
- [49] Dehui Wei, Jiao Zhang, Jialin Li, et al. 2024. Swarm: Cost-Efficient Video Content Distribution with a Peer-to-Peer System. arXiv:2401.15839 [cs.NI]
- [50] Rui-Xiao Zhang, Haiping Wang, Shu Shi, Xiaofei Pang, Yajie Peng, Zhichen Xue, and Jiangchuan Liu. 2024. Enhancing Resource Management of the World’s Largest PCDN System for On-Demand Video Streaming. In *Proc. of USENIX ATC*.
- [51] Shangyue Zhu, Theo Karagioules, Emir Halepovic, Alamin Mohammed, and Aaron D Striegel. 2022. Swipe along: a measurement study of short video services. In *Proc. of MMSys*. 123–135. doi:10.1145/3524273.3528177

A Appendix

A.1 Ethics

We believe this work poses minimal ethical risk. Consistent with the Menlo Report principles [16], all experiments used researcher-controlled accounts, collected no data from real users, generated only low-volume, user-like requests, and confined failure injection to our own clients. We did not target any particular user population, attempt to bypass platform safeguards.

A.2 VoD Dataset Summary

As shown in Table 3, the web and mobile pipelines expose different observables. Web measurements provide session-level watched duration but not chunk counts, whereas mobile measurements expose chunk-level retrieval structure but not directly comparable session-duration measurements. Accordingly, “N/A” denotes metrics that are not defined for that pipeline.

A.3 Limitations of VPN-Based Mobile Redirection

We do not use location-validated VPN endpoints for the mobile measurements for two reasons. First, changing the network egress alone is insufficient to reliably control the native client’s behavior. Public troubleshooting guides also suggest that TikTok evaluates signals beyond IP address, such as timezone and language consistency [48]. As discussed in Section 2, our own web experiments further show that TikTok’s delivery logic can be influenced by persistent application-level state (e.g., cookies) in addition to current network location, and controlling analogous state in the mobile app is substantially less tractable than in the browser.

Second, our mobile instrumentation already depends on system-level traffic capture. In practice, mobile platforms typically permit only a single active VPN interface, so combining per-app capture with an independent VPN tunnel for geographic redirection is operationally fragile and difficult to reproduce consistently across regions. We therefore use distributed cloud deployments for geographic diversity while preserving measurement fidelity and experimental control.

A.4 Quantitative Evidence for Cookie-Controlled Assignment

We use cookie manipulation because changing only the VPN endpoint does not alter the assigned CDN serving region. Starting from a U.S.-created account and its cookies, we vary only the VPN endpoint country. Across 12,785 observations with non-U.S. VPN endpoints, the inferred serving region differs from the VPN country in 100% of cases: in Fig. 10, CDN providers still serve from U.S.-based infrastructure rather than the local VPN country. This confirms that the account-origin cookie constrains the serving region. The inferred U.S. serving locations nevertheless remain close to the corresponding VPN endpoint countries, indicating that proximity still affects server choice within that cookie-constrained region.

Table 3. VoD dataset summary used in the workload analysis.

Metric	Web VoD	Mobile VoD
Sessions	~93k	~7k
Unique videos	72,430	7,161
Avg. session duration (s)	30.91	N/A
Avg. chunks per video	N/A	2.11
Captured volume (GiB)	536.85	4.57

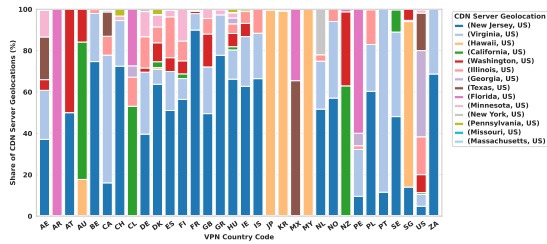


Fig. 10. Serving-location geolocation for a U.S.-created account and its cookies across VPN countries. Colors denote inferred U.S. serving states.

A.5 Controlled-vs.-Residential Validation

To assess whether controlled environments bias delivery-path selection, we perform matched-region comparisons for web (2,260 residential and 6,125 VPN-based videos) and mobile (approximately 1k U.S. sessions across a physical phone, a same-network emulator, and an AWS-hosted emulator). In each case, we compare CDN-provider share first, then condition on the same provider to test for residual differences.

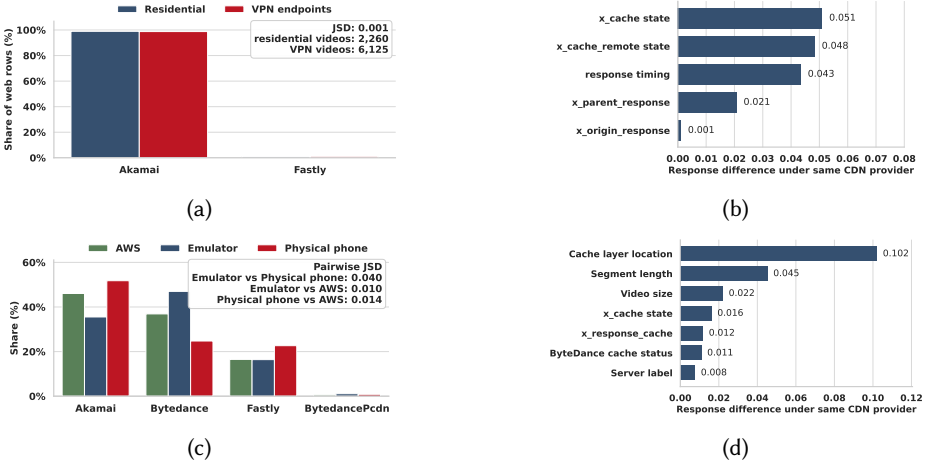


Fig. 11. Controlled-vs.-residential validation. (a, b) Web: CDN-provider share and residual response-level differences between residential and VPN-endpoint sessions. (c, d) Mobile (U.S.): CDN-provider share and residual differences across a physical phone, same-network emulator, and AWS-hosted emulator.

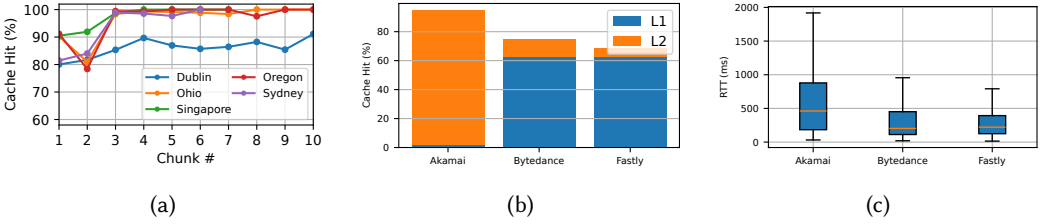


Fig. 12. Mobile VoD cache behavior. (a) Cache-hit rates by chunk position. (b) Cache-hit rates by CDN at U.S. VPs, decomposed by cache layer. (c) RTT distributions by CDN provider.

Fig. 11 shows nearly identical CDN assignment in both environments. For web, provider distributions are almost indistinguishable (Jensen-Shannon divergence 0.001), with only modest residual differences in cache and response-timing signals after conditioning on provider. For mobile, distributions remain close across all pairings (pairwise JSD below 0.05), with the largest variation arising from cache-layer location rather than provider choice. These results indicate that neither VPN-based web collection nor emulator-based mobile deployments materially distort delivery-path selection.

A.6 Cache Behavior Across Mobile Vantage Points

We measure per-chunk cache-hit rates across the five mobile vantage points, classifying a chunk as a cache hit when response headers indicate an L1 or L2 edge hit (using the fields listed in Table 4). Fig. 12a shows cache-hit rate as a function of chunk position; aggregated across all chunks, 89.2% of mobile VoD requests were served as edge hits. At the U.S. vantage points, Akamai's aggregate cache-hit rate exceeds ByteDance's (Fig. 12b), yet average retrieval latency is generally lower for ByteDance (Fig. 12c): Akamai attains 94.6% L1+L2 hit rate but 98% of hits are L2 (regional), whereas ByteDance attains 75% hit rate with 84% served directly from L1. ByteDance's higher L1 residency offsets its lower aggregate hit rate.

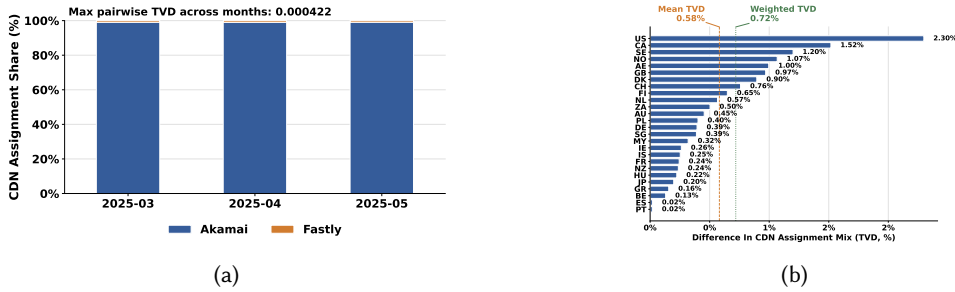


Fig. 13. Robustness checks for web VoD CDN assignment. (a) Provider shares over March–May 2025. (b) Cross-VPN-provider comparison for 29 overlapping countries.

Table 4. HTTP header fields used for CDN identification.

Header Field	Type	Description
Host	Request	Target CDN domain (e.g., *.tiktokcdn.com).
X-ReqType*	Request	Video play type (e.g., preload, play).
x-response-cache	Response	TikTok-specific cache status (e.g., edge_hit).
X-Cache	Response	Cache status (e.g., HIT, MISS) or detailed codes (e.g., TCP_HIT from ...akamaitechnologies.com).
Via*	Response	Cache-layer locations (e.g., clt1a-miss, ord2a-hit).
X-Kfc-Cachekey*†	Response	Cache key (e.g., http://v45.tiktokcdn.com/video/tos/ueast5/tos-useast5-pve-0068-tx/object_id/).
x-parent-response-time	Response	Parent-server response time in seconds (e.g., 12, xxx.xxx.xxx.xxx).
x-origin-response-time	Response	Origin-server response time in seconds (e.g., 34, xxx.xxx.xxx.xxx).
Content-type	Response	Response content type (e.g., video/mp4, audio/mpeg).
X-Bdcdn-Cache-Status†	Response	ByteDance CDN cache status across internal layers (e.g., TCP_HIT, TCP_MISS).
X-Cache-Remote	Response	Upstream cache (parent/shield) status, commonly used by Akamai.
X-Served-By	Response	Fastly cache node serving the request (e.g., cache-iad-...).
x-77-cache	Response	CDN77-specific cache status (e.g., HIT, MISS).
x-77-pop	Response	CDN77 identifier for the L1 PoP (e.g., dublinIE).
X-ID	Response	G-Core Labs identifier for the L1 PoP (e.g., sy4-hw-edge-gc16).
X-ID-Shield	Response	G-Core Labs identifier for the L2 (parent/shield) PoP, following the X-ID format.

* denotes fields specific to mobile HTTP headers., † denotes fields specific to ByteDance CDN.

A.7 Robustness Checks for Web CDN Assignment

We perform two robustness checks for web CDN assignment. First, the aggregate provider mix remains stable over March–May 2025 (Fig. 13a): Akamai stays between 99.033% and 99.075% of web VoD requests, with the largest month-to-month total variation distance (TVD) at 0.000422. Second, comparing 29 overlapping countries across the NordVPN and Surfshark datasets (Fig. 13b) yields mean TVD 0.005817 and maximum 0.022984, consistent with CDN assignment being driven by exit-point geography rather than VPN provider identity.

A.8 CDN Attribution and Provider Mapping

Table 4 lists the HTTP header fields used for CDN attribution in both web and mobile VoD measurements. Fig. 14 shows the fraction of requests attributed by each signal class across web and mobile datasets. Table 6 summarizes the VoD SNI-to-provider mapping, and Table 7 the underlying-ASN breakdown for each live CDN provider in Fig. 5a. For live streaming, CNAMEs follow a regular structure: ByteDance traffic is identified by `bytefcdn-{oversea, ttpu, ttpus}` suffixes across three regional domains, while third-party providers (Wangsu, CDN77, G-Core, Tencent, RocketCDN, EdgeNext) each use distinct suffixes (e.g., `cdn20.com`, `cdn77.org`, `gcdn.co`). QUIC adoption for media delivery is highly asymmetric (Fig. 15): most ByteDance-served U.S. video payload uses QUIC (Ohio: 84%, Oregon: 69.5%), but adoption drops sharply elsewhere; other CDNs deliver almost exclusively over TCP/TLS. Transport selection remains stable under failure injections.

Table 5. Per-country distribution of web VoD fallback taxonomy

Country	Loop	Hier.	Intra	Linear	Total
US	1782	871	51	917	3621
SG	435	641	115	728	1919
FR	303	294	107	1206	1910
CH	258	279	16	1290	1843
DK	100	50	5	1675	1830
SE	199	635	23	964	1821
BE	41	23	0	1748	1812
NL	0	77	0	1729	1806
DE	222	1173	43	352	1790
PT	769	464	96	451	1780
HU	39	32	6	1698	1775
JP	326	747	20	666	1759
ES	252	572	89	845	1758
KR	37	1334	0	363	1734
MY	193	260	30	1250	1733
PL	238	138	38	1311	1725
NO	86	934	47	652	1719
FI	64	1324	0	327	1715
AR	115	271	16	1306	1708
MX	41	981	0	685	1707
GR	8	460	0	1236	1704
CA	625	665	117	264	1671
AE	496	328	169	675	1668
GB	180	189	10	1288	1667
IE	32	21	21	1582	1656
ZA	76	978	76	522	1652
NZ	24	344	0	1265	1633
AU	145	343	40	1095	1623
PE	460	444	47	650	1601
CL	59	411	0	1056	1526

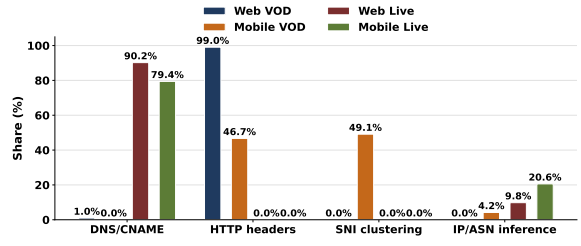


Fig. 14. CDN attribution signal coverage across web and mobile datasets.

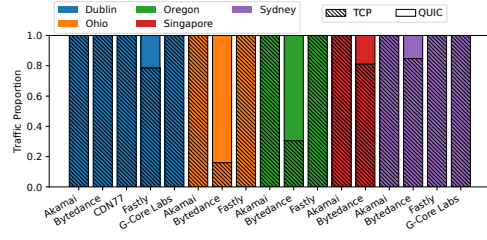


Fig. 15. QUIC adoption in TikTok mobile VoD.

Table 6. TikTok VoD SNI-to-CDN mapping. Providers are identified from the SNI prefixes and domains observed in our VoD measurements.

SNI		CDN	
Prefixes	Domain	Hint	Provider
v16-cla, v16m,	tiktokcdn-us.com	.edgesuite.net	Akamai
v16m-default		.akamaized.net	
v19-cla, v19		.fastly.net	
v45-p, v45		.ttoverseas.net	
v16, v16m, v16-cla	tiktokcdn.com	.edgesuite.net	Akamai
v19, v19-cla		.akamaized.net	
v31-sg, v31		.fastly.net	
v45-p, v45-us		.ttoverseas.net	
v58		.gcdn.co	G-Core Labs
v15m	tiktokcdn-eu.com	.cloudfront.net	CloudFront
v16m		.edgesuite.net	Akamai
v19		.akamaized.net	
v45-p		.fastly.net	Fastly
v58		.ttoverseas.net	ByteDance
v77		.gcdn.co	G-Core Labs
		.rsc.cdn77.org	CDN77

Table 7. Underlying ASNs for each CDN provider. Corporate suffixes are omitted for brevity.

Organization	CDN Hints	Underlying ASNs (Traffic Share)
ByteDance	bytefcdn-oversea	Zenlayer (27.2%), Bedge (23.9%), Akamai (21.9%), CDN77 (20.4%), ByteDance (4.7%), Kaopu (1.8%)
ByteDance	bytefcdn-ttpeu	CDN77 (61.9%), ByteDance (21.3%), Akamai (16.8%)
ByteDance	bytefcdn-ttput	ByteDance (100%)
Wangsu	cdn20	Metaverse (57.1%), IENTC (28.6%), Akamai (14.3%)
Wangsu	wsdvs	Metaverse (70.7%), IENTC (7.2%), Red Cientifica (5.3%), Angola Cables (4.3%), Binariang (2.7%), Akamai (2.5%), Others ($\approx 7\%$)
CDN77	cdn77	CDN77 (100%)
G-Core	gcdn	G-Core Labs (100%)
RocketCDN	rocket-cdn	Bedge (59.5%), Zenlayer (11.4%), Sakura (9.3%), CDSC (7.8%), Tata Comm. (6.7%), TE-AS (3.4%), Others ($< 2\%$)
Tencent	ovscdns	ACE (99.9%), Tencent (0.1%)
BaishanCloud /EdgeNext	spbycdn	Edgenext (45.8%), Zenlayer (38.9%), TM Tech (15.3%)

A.9 Per-Country Fallback Taxonomy Counts

Table 5 reports per-country counts for the four web VoD fallback classes introduced in Section 5.1.

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